The examination is being carried out on the **following application documents**

Description, Pages

1-32	filed in electronic form on	25-01-2024

Claims, Numbers

1-15	filed in electronic form on	25-01-2024

Drawings, Sheets

1/8-8/8	as published

1 Prior Art

Reference is made to the following documents:

D1 US 2020/310461 A1 (KAUFMAN DAVID [US] ET AL) 1 October 2020 (2020-10-01)

D2 AU 2021 201 801 A1 (WAYMO LLC [US]) 15 April 2021 (2021-04-15)

2 Inventive Step (Articles 52(1) and 56 EPC)

2.1 Independent Claim 1

D1 Discloses:

A method of determining a stopping location for a vehicle before picking up a passenger, the method comprising:
by one or more processors, at least one of which is a component of an autonomous vehicle (AV) ("*the server computing system is configured to instruct an autonomous vehicle to pickup a person and transport the person to a desired destination of the person.*", **§0054**):
receiving a ride service request that has been transmitted from a ride service application on a passenger electronic device ("*at 504 the computing system receives a ride request from a mobile computing device, wherein the ride request comprises an identifier of the person, a pickup location, and a destination.*", **§0054** and **Fig. 5**);
determining a pickup area for the ride service request, along with a loading point

that is within a pickup area ("*the person can specify a pickup location for the ride or the pickup location for the ride can be determined based on a current location of the mobile computing device.*", **§0002**);
determining a route from a starting location of the AV to the pickup area;
navigating the AV along the route toward the pickup area ("*the AV coordination component 138 determines a route toward the pickup location of the person*", **§0032**);
before the AV reaches the pickup area, determining whether the one or more processors have received a departure confirmation from the passenger electronic device, wherein the departure confirmation comprises data confirming that coordinates of the passenger electronic device correspond to the loading point (*obvious implementation; the AV can reach the pick-up area at any point without prior confirmation from the passenger*); and
~~if the one or more processors have received the departure confirmation, causing the AV to navigate into the pickup area and stop at the loading point, otherwise causing the AV to either (a) navigate to an intermediate stopping location before reaching the pickup area or (b) pass through the pickup area.~~

The subject-matter of claim **1** differs from what is disclosed in **D1** in that:

"*If the one or more processors have received the departure confirmation, causing the AV to navigate into the pickup area and stop at the loading point, otherwise causing the AV to either (a) navigate to an intermediate stopping location before reaching the pickup area or (b) pass through the pickup area.*"

Although not explicitly disclosed in **D1**, the possibilities of stopping before the pickup point or bypassing it are considered as straightforward alternatives that the skilled person would have implemented without applying inventive step. See also in **D2**, **Fig. 10-16** where the driver has the option to stop on a waypoint before reaching the destination.

Furthermore, the distinguishing feature does not contribute to the technical character of the invention; driving to the pick-up area only after receiving confirmation that the passenger is still on that location is considered by the skilled person as part of requirement specification.

Thus, the distinguishing feature cannot support the presence of an inventive step (Guidelines G-VII 5.4).

Thus, the subject-matter of claim **1** of the present application cannot be considered to involve an inventive step (Articles 52(1) and 56 EPC).

2.2 Independent Claim 10

The same reasoning applies mutatis mutandis for independent claim **10** since it contains the same features as claim **1**.

Thus, the subject-matter of claim **10** of the present application cannot be considered to involve an inventive step (Articles 52(1) and 56 EPC).

2.3 Independent Claim 13

D1 Discloses:

A method of determining a stopping location for a vehicle before picking up a passenger, the method comprising:
by one or more processors, at least one of which is a component of an autonomous vehicle (AV) ("*the server computing system is configured to instruct an autonomous vehicle to pickup a person and transport the person to a desired destination of the person.*", **§0054**):
receiving a ride service request that has been transmitted from a ride service application on a passenger electronic device ("*at 504 the computing system receives a ride request from a mobile computing device, wherein the ride request comprises an identifier of the person, a pickup location, and a destination.*", **§0054** and **Fig. 5**);
determining a pickup area for the ride service request, along with a loading point that is within a pickup area ("*the person can specify a pickup location for the ride or the pickup location for the ride can be determined based on a current location of the mobile computing device.*", **§0002**);
determining a route from a starting location of the AV to the pickup area ("*the AV coordination component 138 determines a route toward the pickup location of the person*", **§0032**);
determining an intermediate stopping location as a location that (a) is along the route between a current location of the AV and the loading point, and (b) satisfies each of a plurality of permissible stopping location criteria (*implicit characteristic*);
navigating the AV along the route toward the pickup area ("*the autonomous vehicle navigates to a destination included in the ride information received*", **§0059** and **Fig. 7**);
before the AV reaches the pickup area, determining whether the one or more

processors have received a departure confirmation from the passenger electronic device, wherein the departure confirmation comprises data confirming that coordinates of the passenger electronic device correspond to the loading point (*implicit characteristic*); and

in response to determining that the system has received the departure confirmation:

causing the AV to navigate toward the pickup area (**Fig. 7**; Steps 704-712),

~~identifying one or more objects located proximate to the loading point,~~

~~measuring a length of a stop zone that includes the loading point and that is free from obstruction,~~

~~using a current speed of the AV and the measured length of the stop zone to determine whether the AV can non-suddenly stop within the stop zone while satisfying each of the plurality of permissible stopping location criteria, and~~

~~stopping at the loading point to perform the loading operation if the system determines that the AV can non-suddenly stop within the stop zone while satisfying each of the plurality of permissible stopping location criteria, otherwise navigating the AV to the intermediate stopping location and notifying the passenger electronic device that the loading operation will be performed at the intermediate stopping location.~~

The subject-matter of claim **1** differs from what is disclosed in **D1** in that:

"*identifying one or more objects located proximate to the loading point,*

measuring a length of a stop zone that includes the loading point and that is free from obstruction,

using a current speed of the AV and the measured length of the stop zone to determine whether the AV can non-suddenly stop within the stop zone while satisfying each of the plurality of permissible stopping location criteria, and

stopping at the loading point to perform the loading operation if the system determines that the AV can non-suddenly stop within the stop zone while satisfying each of the plurality of permissible stopping location criteria, otherwise navigating the AV to the intermediate stopping location and notifying the passenger electronic device that the loading operation will be performed at the intermediate stopping location."

The technical effect is then that the AV stops at a allowable location for a pick-up/drop-off.

The problem to be solved may be regarded as improving safety.

Prior art document **D2** discloses:

"*The computing device 110 may control the direction and speed of the vehicle by controlling various components. By way of example, computing device 110 may navigate the vehicle to a destination location completely autonomously using data from the map information and navigation system 168. Computer 110 may use the positioning system 170 to determine the vehicle's location and perception system 172 to detect and respond to objects when needed to reach the location safely*", **§00331**

and

"*during a fly-by pickup, rather than pulling into a parking area and waiting for the passenger, the vehicle may simply double park, for instance, stopping in a lane when safe to do so and waiting only a few seconds for the passenger to get into the car, close the door and fasten his or her seatbelt. If this occurs, the fly-by pickup has been successful. If not, the vehicle may have to pull away and continue towards the original pickup location. In some cases, the vehicle and passenger may need to rearrange a new pickup location after an unsuccessful fly-by pickup attempt.*", **§0062**.

Thus, the subject-matter of claim **13** of the present application cannot be considered to involve an inventive step (Articles 52(1) and 56 EPC).

2.4 Independent Claim 15

The same reasoning applies mutatis mutandis for independent claim **15** since it contains the same features as claims **1** and **13**.

Thus, the subject-matter of claim **15** of the present application cannot be considered to involve an inventive step (Articles 52(1) and 56 EPC).

2.5 Dependent Claims 2-9, 11-12 and 14

Claim 2: See in **D2** in **§00651-00661** where user input is requested for identifying the suitable pick-up point.

Claim 3: See in **D2** in **§00681** where the pick-up location is proximate to the user device.

Claims 4 and 5: See in **D2** in **§0062** where the pick-up location is in proximity to the pick-up point identified by the user but before that point and the user is notified about the new location that is safer.

Claims 6-8 and 14: See in **D2** in **§0062** where the pick-up location is in proximity to the pick-up point identified by the user but before that point and the user is notified about the new location that is safer. The obstacle being another vehicle is considered by the skilled person as a mere alternative of any possible obstacle.

Claim 9: In case of not having received a departure confirmation, stopping in a safe pick-up location in the proximity of the pick-up point is considered as a straightforward alternative that the skilled person would have implemented without applying inventive skill.

Claims 11 and 12: See in **D2** in **§0062** where the pick-up location is in proximity to the pick-up point identified by the user but before that point and the user is notified about the new location that is safer.

Thus, the subject-matter of claims **2-9, 11-12** and **14** of the present application cannot be considered to involve an inventive step (Articles 52(1) and 56 EPC).

3 Procedure

3.1 To meet the requirements of Rule 42(1)(b) EPC, **D1** should be identified in the description and the relevant background art disclosed therein should be briefly discussed.